

WEEK 3



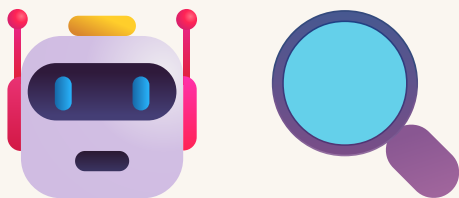
AND Conditions

Two things true at once

두 조건이 모두 참일 때만 (영어코딩)

BIG PICTURE

Smarter corners



With **and**, the agent acts only when **both** checks pass.

둘 다 참일 때만 행동해요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What does **and** mean?

A **and** B is true **only when both** A and B are true.

```
wall ahead and wall on right  
# True only at a dead-end corner
```

One false → the whole thing is false.

하나라도 거짓이면 전체가 거짓

CONCEPT

The truth of and

true and true

✓ true

true and false

✗ false

false and true

✗ false

false and false

✗ false

둘 다 참일 때만 참

CODE

Detect front AND right

```
if agent.detect(BLOCK, FORWARD) and \  
    agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)
```

Both walls present → it's a corner → turn left.

앞과 오른쪽 둘 다 벽이면 왼쪽으로

WATCH IT!

Both must be true

front  + right 

front  + right open

front and right both walls?

✓ true → agent.turn(LEFT_TURN)

✗ false → do nothing

▸ Agent does

CODE

Use it in the maze loop

```
while not agent.detect(BLOCK, DOWN):  
    if agent.detect(BLOCK, FORWARD) and \  
        agent.detect(BLOCK, RIGHT):  
        agent.turn(LEFT_TURN)  
    else:  
        agent.move(FORWARD, 1)
```

Turn left only at true corners; otherwise keep moving.

PREDICT 🤖

Does the agent turn?

Predict first — then click reveal

```
if agent.detect(BLOCK, FORWARD) and \
    agent.detect(BLOCK, RIGHT):
    agent.turn(LEFT_TURN)
# wall ahead, but RIGHT is open
```

Reveal output 🔍 reveal

VOCAB

New words today

and true only when both conditions are true

condition a yes/no test the code checks

both the two checks together, not just one

dead-end a corner blocked on two sides

그리고

조건

둘 다

막다른 길



DEBUG

What's wrong with this code?

Goal: turn left only at a front+right corner.

buggy

```
if agent.detect(BLOCK, FORWARD) and \  
    agent.detect(BLOCK, FORWARD):  
    agent.turn(LEFT_TURN)
```

Think first — then click next.



DEBUG

The bug

Bug: both checks look `FORWARD`, so it turns at *any* wall and spins too early. The two checks must be *different* — front **and** right.

fixed

```
if agent.detect(BLOCK, FORWARD) and \  
    agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)
```

YOUR TURN!

Solve with AND

- ✓ Open the Week 3 world + Code Builder
- ✓ Use at least one **and** condition
- ✓ Turn only at true corners (front + side)
- ✓ Reach the end of the maze
- ✓ Send your walkthrough video on KakaoTalk



QUIZ

QUESTION 1 OF 3

When is **A and B** true?

When at least one is true.

Only when both A and B are true.

Always.

When both are false.



QUIZ

QUESTION 2 OF 3

`detect(FORWARD)` and `detect(RIGHT)` — wall ahead, right open. Result?

It causes an error.

False — both must be true, and the right is open.

It turns left anyway.

True — one wall is enough.



QUIZ

QUESTION 3 OF 3

To find a dead-end corner, the two `detect` calls should...

Be removed entirely.

Both look DOWN.

Look in two different directions, like FORWARD and RIGHT.

Both look FORWARD.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



AND unlocked!

Next: add OR for smarter choices.

다음 주: OR로 더 똑똑한 선택